Jaszmine DeFranco

jaszmine.defranco@gmail.com | 614-769-2255 | jaszmine-defranco-portfolio.vercel.app

Education

The University of Texas at Dallas — December 2025
Bachelor of Science in Computer Science — *Richardson, TX*

Experience

Undergraduate Researcher — May 2025 – Dec 2025
The University of Texas at Dallas - Computer Science Dept. — *Richardson, TX*

- Optimized entanglement-assisted quantum error-correcting codes by reducing required entangled bits, improving resource efficiency in stabilizer-based systems; presented findings at UT Dallas SPUR Symposium 2025.
- Proved a novel qubit-to-ebit transformation preserving error correction capability.

Communications Specialist — Nov 2024 – Dec 2025
The University of Texas at Dallas - Art Sci Lab — *Richardson, TX*

- Led data-driven marketing campaigns, increasing research & event engagement through targeted content.
- Managed end-to-end event logistics & analytics using Meta Business Suite & Power BI to optimize performance.

Software Engineering Intern — May 2023 – Aug 2023
bp; Qubit x Qubit (Early Quantum Career Immersion Program) — *Houston, TX*

- Developed hybrid classical-quantum optimization approaches and solutions using mixed integer programming and quantum unconstrained binary optimization (QUBO) models on NISQ-era systems.
- Collaborated with cross-functional teams through internal COE's and external partners to evaluate applications of quantum computing, AI/LLMs, renewables, & robotics for applications of interest (e.g., energy market operation).

Projects

Tweet Analysis for Disaster Crisis Response | *Capstone at UTD* — Sep 2025 – Dec 2025

- Built a full-stack near real-time crisis detection system processing 25k+ social media posts with over 90% accuracy in classification using LLMs, Bluesky API, React, Next.js, TypeScript, Supabase, PostgreSql, and Hugging Face.
- Led data science development via Jira; deployed scalable architecture using modern web and cloud technologies.

SPOTR | *NASA Micro-g NExT* — Jan 2024 – Jun 2024

- Developed a Search & Rescue Platform for Optical Target Recognition (SPOTR) that utilized a custom-trained YOLO object detection model to identify items of interest for open ocean search and rescue scenarios.
- Utilized a Jetson Orin Nano, PyQT5 GUI, Yolov8 CNN.

Autism Treatment Center App | *Engineering Projects in Community Service* — Sep 2022 – Dec 2022

- Developed a full-stack data collection application to digitize and manage 10+ year's worth of behavioral data, improving accessibility & operational efficiency; exercised engineering best practices (version control, documentation, client liasoning, technical report writing).
- Processed and structured large datasets for reliable access across stakeholders; modernized legacy systems using MERN stack and containerization via Docker, Postman, Auth0, React, TypeScript, and Arch Linux.

Leadership

Facilities Lead | *UTDesign Makerspace* — Jan 2022 – Jan 2025

- Oversaw maintenance & wellbeing of various equipment, tools & rooms; creation of new training, & training new members; documenting critical changes through our discord server.

Simulation Lead | *UTD AIAAA; Competition Rocketry Team* — Jun 2023 – Jun 2024

- Led flight simulation team in designing, building, & testing 2 high-powered rockets (L2 & L3), conducting 6-DOF aerodynamic simulations to optimize flight stability, performance, & safety.
- Competed in the Spaceport America Cup (10k COTS), launching & recovering the L3 rocket at near-target altitudes, achieving mission success.

Technical Skills

Languages: Python, C++, Java, TypeScript, SQL, C, HTML, CSS, JavaScript, Bash
Frameworks: React, Node.js, Next.js, Express.js, Vue.js, Tailwind
Database & Cloud: PostgreSQL, Supabase, Postman, Auth0, MongoDB
Developer Tools: Git, Docker, Google Cloud Platform, CI/CD (GitHub Actions), Figma, Jira/Agile/Scrum
Libraries & Analysis: pandas, NumPy, Matplotlib, Scikit-Learn, OpenCV, MS Excel, Qiskit, Pennylane